

Network Hardware

CSC311 – FALL 2021

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TCP/IP Layers and Devices

Application & Transport Layers (Message and Segment)

- Hosts

Network Layer (Datagram)

- Hosts and Routers (Also Level-3 Switches)

Link Layer (Frame)

- Hosts, Routers, and L2 – Switches

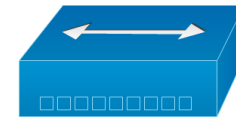
Physical

- All the above

Repeaters, Hubs, and Bridges

Hubs

- Physical Layer
- “Dumb” Devices
- Packets arriving at a hub’s input port get sent to every other interface.



Hub

Bridges

- Link Layer (Layer 2)
- Segment networks based on MAC Addresses



Bridge

Repeaters

- Physical Layer
- Used to regenerate signal when going long distances
 - Ethernet has a limit of about 100 meters



Repeater

Images sourced from: <https://www.conceptdraw.com/>

Switches (Section 6.4.3)

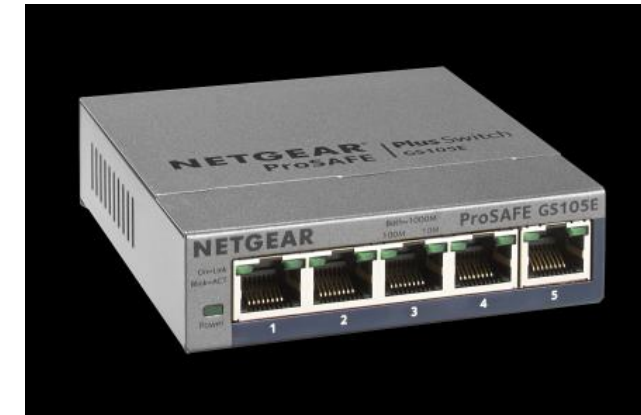
Switches

- Typically operate at Layer 2
- Switches keep track of which device is connected to which interface
- Switch (Forwarding) Tables

Packets arriving at a Switch, are only forwarded to their intended target.

Mac Addresses –

- MAC – Media Access Control
- A “unique” identifier assigned to every network interface
- 4C-CC-6A-18-29-8A



Netgear 5-Port Switch

Switch Table (Simplified)

MAC Address	Interface	Time
4C-CC-6A-18-29-8A	1	3:40
AA-DD-EE-FF-11-22	2	3:41
....		...
0C-12-1E-21-2D-33	10	1:00

- MAC Address – The MAC address of the machine connected to that interface
- Interface - Which interface (or port) leads to the MAC address
- Time – When the entry was placed in the table

Self-Learning

Switches build tables automatically, dynamically, or autonomously

1. Switch Table is Initially Empty
2. For each frame (Which layer is this?) that is received, the switch will store the **SOURCE** MAC address and current time.
 1. Once every host in the LAN sends a frame, each host will be recorded.
3. Switch deletes an address in the table if no frames are received with that address as the source address after some period of time (aging time). If the PC is replaced with another, that will result in a new row in the table, and eventually the old will be purged from the table.

Routers (Section 4.2)

Interact with Layer 3 (Networking)

Input Ports

Switching Fabric

Output Ports

Routing Processor

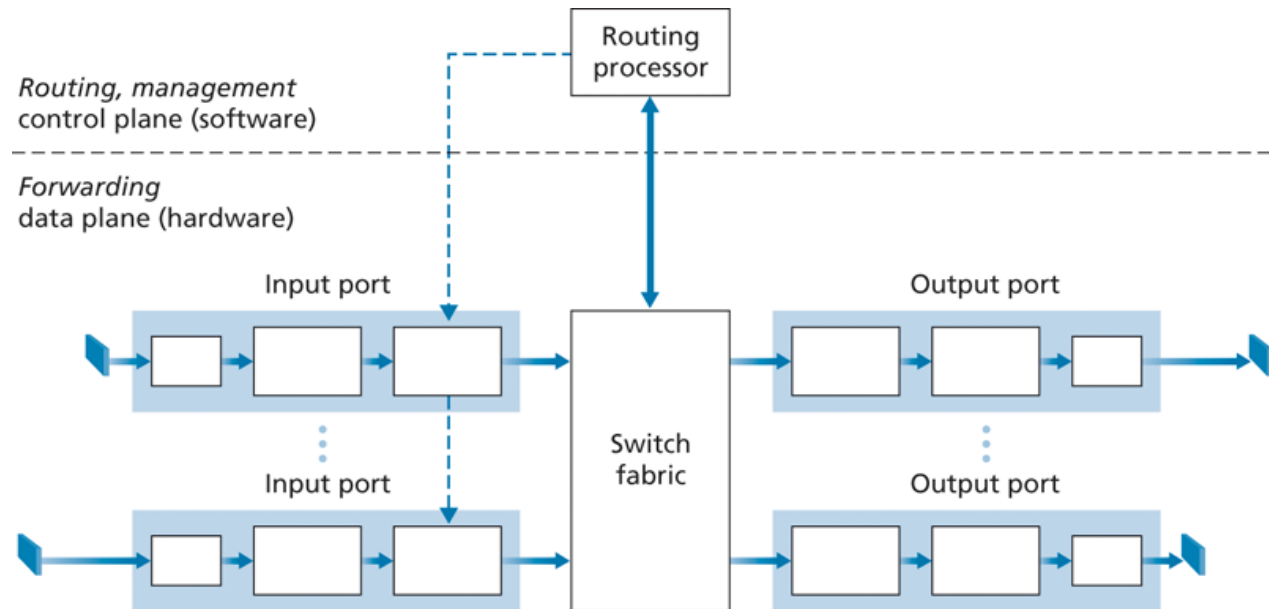


Figure 4.4 (pg 311)